From Bridge Tables to the Negotiation Table: Cross-Domain Behavioral Consistency of Game-Derived Decision-Making Profiles

Anna Ben-Shushan[1], Jari Hämäläinen[1,2], Pekka Neittaanmäki[1], and Nezer Jacob Zaidenberg[3]

[1]Department of Computational Engineering, LUT School of Engineering Sciences, LUT University, Lahti, Finland

[2]Slovak University of Technology, Bratislava, Slovakia

[3]Ariel University, Ariel, Israel

Corresponding author: anna.ben.shushan@student.lut.fi

Draft, September 2026

Prepared for the Springer OP80 volume *Recent Advances on Scientific Computing, AI, and Other Software Tools*

Abstract

Business negotiation is one of the most consequential forms of decision-making under uncertainty, yet it is difficult to study empirically: real negotiations are private, and no public dataset links a long record of a person's actual decisions to how they negotiate. We propose contract bridge as a measurable laboratory. Bridge bidding is a structured sequence of decisions under incomplete information in which every action is recorded and scored, and the same player is observed across hundreds of comparable decision problems. From 149,208 board-level records of five European championships (2016–2025) we find, first, that the behavioral features of elite players show no clearly separated clusters under any of five preprocessing configurations; styles are better described as a *continuum*. We therefore identify the extremes of each behavioral axis as four decision-making profiles; a held-out test shows that these extremes are only weakly stable within a player's own record, so we treat the profiles as extreme records rather than as fixed traits. We turn each profile into an autonomous software agent whose behavior is grounded in the skills that a language model reads from the actual deals of players in that profile. The agents then play bridge and negotiate simulated business deals against fixed counterparts. Two findings emerge. Whether a profile wins in one domain says little about whether it wins in the other: winning depends on the opponent and on the rules of the game. How a profile behaves, however, is consistent across the two domains: agents of profiles whose players took more risks at the bridge table make bolder demands at the negotiation table, and a control experiment that swaps the underlying skills between profiles reverses this relationship, indicating that the behavior is carried by the game-derived skills rather than by the profile's name. These findings concern profile-conditioned software agents, not the human players themselves, and with five profiles they are effect-size indications rather than statistically significant results. The study is a feasibility result rather than a proof, and all data, code, and outputs are released so that every number in the paper can be reproduced.

Keywords: decision-making under uncertainty; contract bridge; negotiation; large language model agents; behavioral profiling; cross-domain transfer; reproducibility.

1 Introduction

Negotiation, over prices, alliances, contracts, and platforms, is a sequence of decisions under uncertainty with conflicting objectives: demand more and risk the deal, or concede and lose surplus. Despite its economic weight, negotiation behavior is hard to study at the level of the *individual decision-maker*. Real negotiations are private and rarely recorded; the transcripts that are public (teaching cases, mediation examples) are edited and anonymized; and, most importantly, no public dataset links the *same person's* decision-making style, as revealed by a long record of their actual decisions, to their observed negotiation behavior. The closest resource, the CaSiNo corpus, pairs negotiation dialogues with questionnaire-based personality scores rather than with behavioral records (Sec. 2).

Contract bridge offers a way around this measurement problem. Bridge bidding is a formal negotiation-like protocol: partners exchange information through a constrained language of bids while opponents contest the auction, all under incomplete information (each player sees only 13 of 52 cards). Unlike real negotiations, *every* decision is recorded, timestamped to a specific deal, and objectively scored, and elite players are observed over hundreds of comparable decision problems. Bridge has long served as a benchmark for AI under imperfect information [1, 2]; here we use it differently: not to play bridge well, but as a *measurable laboratory of individual decision-making style*.

This paper asks a single question:

> *If we derive a player's decision-making profile from their real bridge record and instantiate it as an autonomous agent, does the profile's behavioral signature persist when the agent negotiates in a business setting?*

We answer it with a four-stage pipeline (Secs. 4 to 7) and report one positive result and two instructive negative ones.

The paper makes five contributions, stated here in the order in which the reader meets them. The first is a negative result about the data itself: across five escalating preprocessing configurations and three algorithm families, the behavioral features of elite championship players show no clearly separated cluster structure (Sec. 4.2). We therefore treat style space as a *continuum* for the purpose of profiling; the absence of detected clusters does not by itself prove that the underlying structure is continuous, but it redirects the methodology away from partitioning. The second contribution follows from it: we profile the *tails* of the continuum with a statistically gated rule, under which a player is assigned to a profile only if they lie in the top decile of the profile's defining axis and a one-sided binomial test rejects the population baseline (Sec. 4.3). We also document how expert review exposed a small-sample artefact in our first version and how the gate repairs it, a reusable recipe for profiling rare behaviors. The third contribution is an *evidence-grounded* way of turning a profile into an agent: a large language model (LLM) reads a seeded sample of raw deals from 20 profiled players and writes observable skills with per-deal evidence citations; it is directed to the profile's defining axis but never shown the engineered feature values, so every skill must be verifiable from the cited deals (Sec. 5). Fourth, we build a *cross-domain testbed* in which profile-conditioned agents play bridge and negotiate against fixed, rule-based counterparts; the bridge metric is checked against a zero-intelligence baseline [13] and a double-dummy reference [1], and the shape of the negotiation counterpart is grounded in 5,247 human bargaining dialogues [7] (Sec. 7). The fifth contribution is the principal finding, which concerns the agents and not the human players: the agents' behavioral *style* is consistent across the two domains (rank correlation $\rho_S = +0.80$ over the five profiles, an effect-size indication that cannot reach statistical significance with five points), and an inverse-prompt control that swaps the extracted skills between profiles reverses the relationship ($\rho_S = -0.90$), indicating that the consistency is carried by the skills rather than by the profile label. Whether a profile *wins* in both domains shows no stable relationship

($\rho_S \approx +0.2$ to $+0.5$ depending on the metric): winning depends on the opponent and the payoff structure (Sec. 9).

Throughout, we adhere to two methodological principles: every experimental condition needed to reproduce the study is stated explicitly (Appendix A maps each reported number to a runnable script), and evaluation scenarios are ordered from *trivial*, where the reader can predict which profile should prevail and so builds trust in the instrument, to non-trivial.

The rest of the paper is organized as follows. Section 2 places the work in the literature on bridge AI, negotiating agents, and the psychology of behavioral consistency. Section 3 describes the corpus. The pipeline is then presented in four stages, each a section of its own: Stage 1 (Sec. 4) derives behavioral features from the records and discovers the decision-making profiles; Stage 2 (Sec. 5) has a language model read each profiled player's deals and write down the observable skills; Stage 3 (Sec. 6) builds an agent from each profile; and Stage 4 (Sec. 7) evaluates the agents in bridge and in negotiation. Section 8 describes how the pipeline is engineered as research software. Section 9 reports the results, Section 10 interprets them and states the limitations, and Section 11 concludes with future work. Everything needed to reproduce the results is collected in Appendix A.

2 Related Work

Bridge has long served as an *AI benchmark*. Ginsberg's GIB introduced double-dummy analysis as an objective reference for bridge decisions [1], and Rong et al. trained estimation and policy networks for competitive bidding on about one million deals [2]. These works optimize play strength; we instead treat the human record as behavioral data.

A second line of work builds *game-playing and negotiating agents*. Self-play systems master perfect-information games [3]; Diplomacy work combines language and strategic reasoning and studies honesty and sanctioning in negotiation-like play [4, 5]. End-to-end negotiation dialogue agents were introduced by Lewis et al. [6]; He et al. decoupled strategy from generation and released the Craigslist Bargaining corpus we use for calibration [7]. Human negotiation corpora with per-participant trait measures now exist, notably CaSiNo, a multi-issue corpus with Big-Five and social-value-orientation scores per negotiator [11]. For LLM negotiators, NegotiationArena benchmarks multi-turn agent-vs-agent bargaining and shows that injected behavioral tactics measurably shift payoffs [9]; Abdelnabi et al. contribute multi-party, multi-issue scorable negotiation games [10]; and a 180,000-negotiation competition finds that prompted *warmth* and *dominance* systematically shape outcomes [8], independent evidence that prompt-carried style becomes bargaining behavior. Our contribution is orthogonal to all of these: we do not train or benchmark a negotiator; we test whether a profile derived from *real human decision records* in one domain persists, in an agent, in another domain.

The *zero-intelligence* tradition supplies our null model. Gode and Sunder showed that budget-constrained random traders ("ZI-C") achieve high allocative efficiency, making them the canonical null model for market behavior [13]. We adopt a ZI-C-style bidder as the falsification baseline for our bridge skill metric: a metric that a constrained random agent can top warrants scrutiny before it is used to measure skill.

Our profiling method is a relative of *archetypal analysis*, which represents data by its extreme points rather than by central prototypes [15]; our tail profiling operates on individual, interpretable behavioral axes with an explicit significance gate, rather than fitting a convex-combination factorization. From behavioral economics, prospect theory motivates why payoff structure shapes risk appetite [14], coopetition frames the simultaneous cooperation and competition that bridge partnerships embody [16], and classical negotiation analysis provides the surplus-splitting lens [17, 18].

Whether behavioral dispositions persist across situations, the question of *cross-situational consistency*, is a classical one in personality psychology: consistency emerges reliably only when

behavior is *aggregated* over many observations [30], and stable dispositional signatures coexist with systematic situational variation [31]. For risk attitude specifically, human self-report evidence indicates substantial domain-specificity [32]. These findings calibrate the scope of our claim: we measure the cross-domain consistency of *profile-conditioned agents*; whether the same consistency holds for the underlying human players is a distinct, open question left for future work with trait-annotated human negotiation data (Sec. 11).

Finally, language models are increasingly used as *behavioral simulacra*: conditioned on personas, they can reproduce human-like behavioral distributions [25, 26, 27]. Two validity risks are documented. The first is *tautology*: an agent told it is aggressive will act aggressively. The second is the converse, *non-enactment*: role-playing agents can state a persona's beliefs yet fail to behave accordingly [12], so persona conditioning cannot simply be assumed to work. Our design addresses both empirically: skills are extracted from raw hands under a deal-level evidence requirement (Sec. 5), and an inverse-prompt control swaps skills between identities and measures whether behavior follows (Sec. 9.3), precisely the belief-vs-behavior consistency check this literature calls for.

3 Data: 149,208 Boards from Five European Championships

The corpus was scraped from the public tournament archive of the European Bridge League (EBL) for five championships: Budapest 2016, Ostend 2018, Madeira 2022, Herning 2024, and Poznan 2025. The categories covered are Open, Women, and Senior in 2016 and 2018; Open, Women, Senior, and Mixed in 2022 and 2024; and Open and Mixed in the 2025 archive. The unit of record is one board (deal) played in one room. In team play every board is played twice, in an Open and a Closed room, by different pairs holding the same cards, so each deal yields two independent auctions and results; this duplicate structure is what later gives us an opponent-controlled skill measure (Sec. 7.1). The corpus holds **149,208 board-room records** with 48 fields each and about 2,500 named players (Table 1).

Table 1: Corpus composition and field coverage per championship.

Championship	Rows	Player names	Card holdings	Bidding sequences
Budapest 2016	38,240	100%	56%	0%
Ostend 2018	32,384	100%	53%	0%
Madeira 2022	32,256	100%	51%	0%
Herning 2024	34,048	100%	56%	99.7%
Poznan 2025	12,280	100%	51%	100%
Total	149,208	99.9%	54%	31% (46,230 sequences)

Each record carries four groups of fields: identity (championship, category, match, round, board, room); the four players seated in that room, recovered from the published seating diagrams; the deal itself, that is, the 52 cards by seat, where the EBL published them; and the auction, followed by the contract, the declarer, a doubled marker, the number of tricks taken, and the score. The auction is stored as a string of seat-labeled calls. Each item reads `seat:call`, where the seat is N, E, S, or W and the call is `Pass`, a bid (a level from 1 to 7 followed by the suit: `C` clubs, `D` diamonds, `H` hearts, `S` spades, or `NT` for no trump), or `x` for a double. Calls are listed four to a row starting from West, rows are separated by `|`, and `-` marks a seat whose turn came before the dealer's. One record, taken verbatim from the corpus, reads (dealer East, all vulnerable):

```
W:- N:- E:Pass S:1C | W:1D N:1S E:Pass S:2C |
W:Pass N:3NT E:Pass S:Pass | W:Pass
```

Read from left to right: East passed, South opened one club (`1C`), West overcalled one diamond (`1D`), North bid one spade, East passed, South rebid two clubs, West passed, North jumped to three no trump (`3NT`), and the other three players passed, which ends the auction. This is Board 10 of a Round-5 match in the Mixed category at Herning 2024, Open Room: North played 3NT, took nine tricks, and scored +600. Two quality flags mark whether the auction and the cards are present, and every analysis filters on the flag it needs rather than discarding partial records.

Coverage differs by championship for reasons of publication, not of play. Older EBL microsites do not expose machine-readable auctions, so bidding-dependent features use the 46,230 records of Herning 2024 and Poznan 2025 that carry an auction (31% of the corpus, Table 1; 98 Herning records, 0.3%, were published with a result but without an auction and enter the outcome features only), while contract and outcome features use all 149,208; card holdings are published for 54% of boards. Cleaning is light and mechanical: contract strings are parsed into level, strain, and doubled status; a contract is scored as made when the tricks taken reach the level plus six; and the declarer's name is resolved from seat and room. Three things the corpus does not contain are worth stating, because they bound what the paper can claim: the card-by-card play of each hand, the time taken over each call, and the partnership's bidding system. All data are public tournament records of named professional players published by the EBL; we collect no private or demographic information.

4 Stage 1: Discovering Decision-Making Profiles

4.1 Behavioral Features

For every player we compute per-player rates from two complementary families: *outcome* features, derived from the final contract the player declared (what they committed to), and *process* features, derived from their calls inside the auction (how they got there). A player enters the analysis only with $n_{\text{declared}} \geq 50$ declared boards *and* $n_{\text{bidding}} \geq 50$ boards with auction data, leaving **563 qualifying players**. Interpreting per-player rates only after aggregating ≥ 50 observations follows the aggregation principle of behavioral-stability research [30]: single-situation behavior is noisy, but aggregated tendencies are stable. Sec. 4.4 shows how far 50 boards fall short of that ideal for rare behaviors such as slams.

Table 2: Behavioral features. Family A is computed on declarer boards (all 149K rows); Family B on any seat from full auctions (46K rows). Features with coefficient of variation cv < 0.10 across players (e.g. mean contract level, calls per board) carry almost no between-player signal and are excluded from structure analysis.

Family	Feature	Definition (per-player rate)
A (outcome)	`slam_rate`	declared contracts at level 6–7
A	`partscore_rate`	declared contracts at levels 1–3 (stopping low; a level-based proxy that counts 3NT as a stop)
A	`nt_rate`	declared contracts in no-trump
A	`double_rate`	own declared contracts doubled by opponents
B (process)	`opening_rate`	boards where the player opened the auction
B	`preempt_rate`	openings at level ≥ 2 (obstructive; also counts the strong 2♣ and 2NT openings)
B	`intervention_rate`	entries into the opponents' auction
B	`penalty_double_rate`	boards where the player doubled the opponents[1]

Note the asymmetry: `double_rate` is partly opponent-dependent (being doubled requires the opponents' decision) and is therefore *not* used as a profile-defining axis below; all four defining axes are the player's own actions.

4.2 Finding 1: No Clearly Separated Clusters

To ensure that the absence of cluster structure is not an artefact of inadequate preprocessing, we ran a five-configuration audit before accepting any negative result: (V1) StandardScaler only; (V2) StandardScaler + PCA(3) [37]; (V3a–c) RobustScaler + cv-filter + correlation pruning + Mahalanobis outlier removal ($\alpha = 0.01$) + PCA(5), feeding K-Means [35], a Gaussian mixture with BIC selection [36], and HDBSCAN.

Table 3: Cluster-structure audit. Silhouette follows Rousseeuw [19]; HDBSCAN follows [20].

Config	Algorithm	Best k	Silhouette	HDBSCAN clusters
V1	K-Means	4	∼0.15	0
V2	K-Means (PCA 3)	4	**0.24**	0
V3a	K-Means (full prep.)	2	0.17	n/a
V3b	GMM + BIC	2	0.14	n/a
V3c	HDBSCAN	n/a	n/a	**0**

Three observations converge. First, silhouette never exceeds 0.24 (values near 0 mean points sit on cluster boundaries). Second, the density-based HDBSCAN, free of K-Means' convexity assumptions, labels *all* 563 players as noise in every configuration. Third, *more* aggressive preprocessing makes separation *worse* (V2→V3a: 0.24 → 0.17): outlier removal deletes precisely the extreme players who carry the weak signal. The PCA spectrum decays gradually (PC1 = 24.6%, PC2 = 19.1%, PC3 = 13.5%, …) with no dominant component, which is compatible with a smooth distribution though not a proof of one. We conclude that, under the configurations investigated, the selected features (including the two proxies noted in Table 2) do not separate elite championship players into distinct clusters, a picture consistent with expertise research on maximal adaptation to task constraints among top performers [24]. We therefore treat the style space as a **continuum** for the purpose of profiling, while noting that the absence of detected clusters does not by itself establish continuity. Partitioning such data is the wrong operation; profiling its *tails* is a defensible alternative.

4.3 Extreme-Percentile Profiling with a Significance Gate

Let $x_a(i)$ be player i's rate on defining axis a, with population mean μ_a, standard deviation s_a, and z-score $z_a(i) = (x_a(i) - \mu_a)/s_a$. The four defining axes and their profiles are: `slam_rate` → *Slam Hunter*, `partscore_rate` → *Insurance Player*, `penalty_double_rate` → *Fighter*, and `nt_rate` → *NT Specialist*. Player i is assigned to profile $a^* = \arg\max_a z_a(i)$ iff both gates hold:

$$z_{a^*}(i) \;\geq\; Q_{0.90}\big(z_{a^*}\big) \qquad \text{and} \qquad \Pr\big[X \geq k_i \,\big|\, X \sim \mathrm{Bin}(n_i, \mu_{a^*})\big] < 0.05, \tag{1}$$

where k_i is the player's raw event count and n_i the relevant denominator (declared boards for Family A axes; bidding boards for the Fighter axis). Players failing either gate are *Generalists*, retained as the elite baseline, not discarded. Profile names are human-chosen domain labels for "extreme on axis a"; nothing downstream depends on the name (Sec. 9.3 tests this). The complete per-player assignment, with player names, is released with the data as the file `player_profiles.csv` (Appendix A.2); we do not reproduce names in the text, because Sec. 4.4 shows that the labels describe records in this corpus rather than persons.

[1]The feature counts all doubles (takeout and penalty alike); both are an active choice to contest the auction, so it measures *aggressive doubling*. We keep the historical column name for reproducibility.

The binomial gate exists because of an *expert-review episode* that is worth recording. Our first release (v2.0) used $n_{\text{declared}} \geq 20$ and the percentile gate alone, yielding 807 players and 64 Slam Hunters with a mean slam rate 2.8× baseline. Expert review by a national-level bridge player flagged a rare-event artefact: with a ∼4–6% slam baseline, a player with 20 boards and 2 slams shows a 10% rate whose exact 95% confidence interval, [1.2%, 31.7%] (Clopper–Pearson [38]), is indistinguishable from the baseline. Indeed 41% of v2.0 "Slam Hunters" had fewer than 30 declared boards. Version 2.1 raised the floor to 50 boards and added the binomial gate of Eq. (1): the cohort shrinks to 20 Slam Hunters whose median n_{declared} jumps from 42 to **216** and whose CI, $[6.3\%, 15.2\%]$, clears the baseline. Effect ratios become more conservative (e.g. 2.8× → 1.84×), and every surviving assignment is individually distinguishable from the population rate at $p < 0.05$. We report this revision openly as a reusable methodological recipe: *for low-base-rate behaviors, a percentile cutoff alone manufactures false profiles; gate it with a significance test against the population rate.* The gate is, however, a per-player filter and not a corpus-level test; whether the resulting cohorts are stable behavior is a separate question, which Sec. 4.4 tests out of sample.

Table 4: Final profile assignment (v2.1; 563 players). "Ratio" is the profile's mean on its defining axis over the Generalist mean, measured in sample and therefore large by construction (see Sec. 4.4).

Profile	n (%)	Defining axis	Profile mean	Generalist mean	Ratio	Median n_{declared}
Slam Hunter	20 (3.6%)	`slam_rate`	0.101	0.055	1.84×	216
Insurance Player	21 (3.7%)	`partscore_rate`	0.684	0.570	1.20×	97
Fighter	37 (6.6%)	`penalty_double_rate`	0.131	0.085	1.55×	153
NT Specialist	17 (3.0%)	`nt_rate`	0.385	0.282	1.36×	114
Generalist	468 (83.1%)	(baseline)	n/a	n/a	n/a	117

4.4 Are the Profiles Stable Behavior? An Out-of-Sample Test

Are the profiles stable behavior or a selection artefact? Comparing profile members with Generalists on the defining measure, within the same data, cannot answer this: the members were selected for being extreme on that measure, so large separations are guaranteed by construction. An earlier draft reported such in-sample comparisons (Cohen's d between 2.1 and 4.6); we withdraw them as evidence. The question needs data that played no part in the selection. We therefore split the corpus into two halves that share no board, assigned profiles on one half only, and measured the defining rates on the other half. The split is made at the level of whole matches (a seeded random half of the matches against the rest), which balances dealer, vulnerability, and opponents between the halves. A naive odd/even split by board number is not usable here: in duplicate bridge the dealer rotates with the board number, so one parity is always dealt by North or South and the other by East or West, which biases every seat-dependent rate (we verified that it produces spurious negative reliabilities). A player qualifies only with at least 50 declared and 50 bidding boards in each half, the paper's own threshold, leaving 259 players qualified in both halves.

Table 5 gives the result. On the held-out half, profile members exceed the Generalists by between 0 and 13 percent on their defining axis, with small effect sizes; only the Fighter reaches $p < 0.05$, and the Insurance Player shows no out-of-sample difference at all. Profile membership is unstable: of the players assigned an extreme profile on one half, 4.8% receive the same profile on the other half and 16.7% any extreme profile, and of the published profile members who qualify in a half (54 of 95), 46% are assigned the same profile from that half alone. The underlying per-player rates have low split-half reliability (Pearson r between a player's rate on the two halves: slam 0.07, partscore 0.15, no-trump 0.08, penalty double 0.18,

preempt 0.18, opening 0.23). A multiple-testing argument points the same way: the binomial gate is a per-player filter at $\alpha = 0.05$, so among 563 players about 28 per axis would pass by chance, which is the order of the cohort sizes in Table 4. Each member's record is therefore individually distinguishable from the baseline, but the cohorts are not demonstrated stable traits of the players: they are *extreme records* in this corpus. The 20 Stage-2 source players, whose hands seed the agents, are the high-volume end of the cohorts and fare better: all five Slam Hunters exceed the Generalist mean on both halves of their own boards, as do three of five Fighters and three of five NT Specialists, and two of five Insurance Players (the remaining players lack 50 boards in one half). Fig. 1 shows the in-sample behavioral fingerprints of the full cohorts, which are descriptive only. This result reframes what the rest of the paper can claim: the agents are grounded in real extreme records, and the findings about them are findings about agents, not about stable dispositions of the human players.

Table 5: Out-of-sample validation. Profiles are assigned on one half of the matches and the defining rates measured on the other half (one-sided Mann–Whitney U [22]; Cohen's d [21]). n counts members qualified in both halves; 217 Generalists form the comparison group.

Profile	Defining metric	n	Ratio (held-out)	Cohen's d	p
Slam Hunter	`slam_rate`	12	1.11×	0.27	0.11
Insurance Player	`partscore_rate`	7	1.00×	−0.03	0.43
Fighter	`penalty_double_rate`	17	1.13×	0.42	0.025
NT Specialist	`nt_rate`	6	1.04×	0.26	0.31

Several *known confounds* are retained as limitations. Bridge is a partnership game: part of an elevated slam rate may reflect an aggressive partner rather than the individual. The NT rate partly reflects the pair's bidding system (strong-club systems such as Precision open 1NT on a different point range). Two features are level-based proxies rather than strict bridge categories, and we keep them as computed for reproducibility: `partscore_rate` counts 3NT, a game contract, among the low stops, and `preempt_rate` counts the strong 2♣ and 2NT openings (27.7% of all openings at the two level or higher in this corpus) alongside the obstructive ones. We measured what changes under the strict bridge definitions (partscore: level 1–2, a suit contract at level 3, or 4♣/4♢; preempt: openings at level 2 or higher except 2♣ and 2NT; script in the repository). The Slam Hunter, Fighter, and NT Specialist cohorts are unchanged or nearly so (Jaccard overlap 1.00, 0.95, and 0.94), the bridge-aggression composite of Sec. 9.2 gives the same $\rho_S = +0.80$ with the strict preempt definition, but the Insurance Player cohort is largely a different set of players under the strict partscore definition (overlap 0.14; four of its five Stage-2 source players lose the profile). The Insurance Player is therefore the fragile profile of this study: its signal is the weakest (1.20×), it has no out-of-sample support (Sec. 4.4), and it depends on the proxy definition. The Fighter is methodologically the cleanest profile: doubling is a deliberate individual act, its denominator counts both sides of the table, and its base rate (∼8.5%) narrows confidence intervals quickly.

5 Stage 2: Evidence-Grounded Skill Extraction

Skill extraction runs on a Stage-2 sample of 20 players: five drawn at random (seed 42) from each of the four extreme profiles. For each of them we sample up to 50 boards (deterministic, seed 42) and format them in chunks of 25 boards per LLM call. Each board is rendered as four compact lines, showing the player's 13 cards, the full auction with seats labeled, the final contract, and the numeric result. A real board from the corpus (Herning 2024, Women, Round 19, Board 28) renders as:

```
Board 28 (Dealer West, Vul: N-S)
```

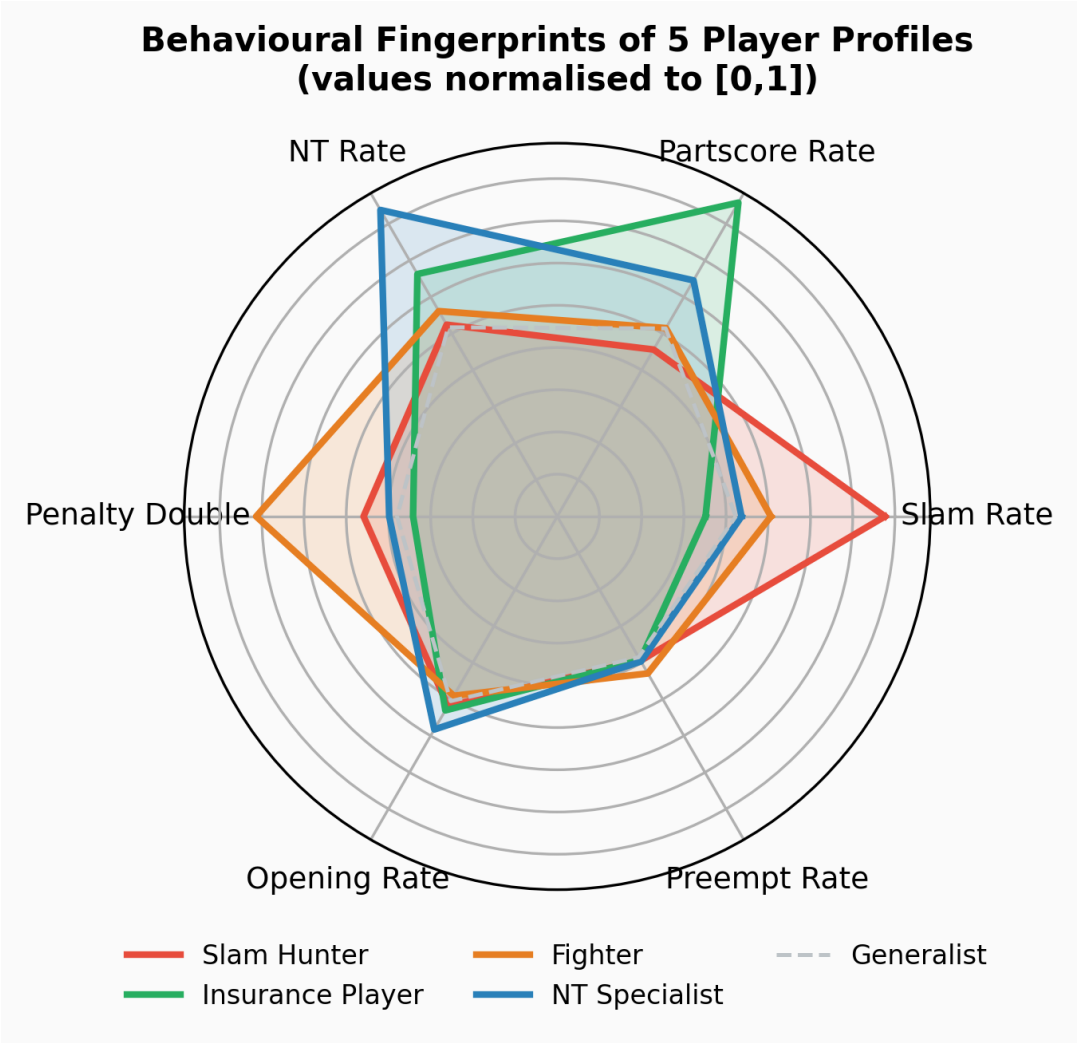


Figure 1: In-sample behavioral fingerprints of the four extreme profiles against the Generalist baseline (each feature min–max normalized to $[0, 1]$ across all players; descriptive only, see Sec. 4.4). Each profile is extreme on its own defining axis, and only mildly deviant elsewhere.

```
You are S (Declarer). Your hand: ♠AK752 ♡6 ♢AKQ ♣QT74
Bidding: W:Pass N:Pass E:Pass S:1S | W:Pass N:4C E:Pass S:4NT |
         W:Pass N:5C E:x S:Pass | W:Pass N:5H E:Pass S:6C |
         W:Pass N:6H E:Pass S:6S | W:Pass N:Pass E:Pass
Result: 6S by S, made 12 tricks, NS +1430
```

The extraction model (`gemini-2.5-flash`, temperature 0.3, JSON-schema output) is instructed to act as a bridge expert and return 3–5 *observable, falsifiable* skills, each with a name, a one-sentence behavioral description, per-board evidence citations, and a confidence grade; personality adjectives are explicitly forbidden. A profile-context block deliberately directs the model's attention to the profile's defining behavioral *axis* (e.g. "this player is in the top decile of slam bidders; at least two skills must concern slam bidding"); without such direction, rare defining behaviors drown under generic competitive-bidding observations. The model never receives the engineered feature values, the z-scores, or any comparative statistics: everything it asserts must cite specific deals as evidence, and Sec. 4.4 reports, independently of any LLM, how far the directed-to behaviors are present in these players' records in and out of sample. Whether downstream behavior follows skill content rather than profile identity is tested separately by the inverse-prompt control (Sec. 9.3). Skills are aggregated across chunks and players

into a per-profile signature by TF–IDF cosine matching with a merge threshold of 0.40 (at 0.30, unrelated skills merged falsely; the threshold sweep is in the repository). TF–IDF is preferred over neural embeddings deliberately: at this scale (dozens of short, domain-specific skill names) lexical overlap is the relevant signal, and the aggregation stays deterministic and auditable, in line with the reproducibility goals. Typical extracted skills: *"uses Roman Key Card Blackwood to drive to slam with fitting hands"* (Slam Hunter); *"doubles opponents' partscores for penalty with trump stacks"* (Fighter); *"declines game invitations with borderline values"* (Insurance Player).

This design carries the first half of the paper's anti-tautology burden: the downstream agents are conditioned on *what the player demonstrably did with the cards*, not on adjectives we invented. The second half, that agent behavior follows this content rather than the profile's name, is carried by the inverse-prompt control (Sec. 9.3).

6 Stage 3: Profile-Conditioned Agents

Each of the five agents (four profiles plus the Generalist control) is an LLM (`gemini-2.5-flash`) whose system prompt is assembled from: (i) an identity line naming the profile and its one-line essence; (ii) the Stage-2 skill list, the only content derived from the players' actual records; (iii) domain rule constraints; and (iv) a mandatory JSON output schema with a free-text rationale field. The identity line does name the profile; whether measured behavior is driven by that name or by the skill content is exactly what the inverse-prompt control of Sec. 9.3 tests, by swapping only the skills while leaving every identity line in place. Bridge agents run at temperature 0.3 (later 0.0), negotiation agents at 0.7.

As a *mechanism check*, all five agents received the same two hands under the final prompts (Table 6; the archived outputs are in `experiments/`). The first is a flat 12-point hand (♠KJ95 ♡Q72 ♢KQ4 ♣J83) after the right-hand opponent opens 1♡: a competitive decision with no textbook answer. Here the call tracks the profile. The Insurance Player passes ("the safest option"), as does the Generalist ("not enough for a 1NT overcall, no five-card suit"), while the Fighter overcalls 1♠ on a four-card suit, "prioritizing competitive presence over strict suit-length requirements", and the Slam Hunter and the NT Specialist also act. The second is a 22-point hand with five-card support (♠AKQ72 ♡AK4 ♢A83 ♣Q2) after partner opens 1♠, where the rules of the game leave little room: four of the five agents make the textbook game-forcing raise (Jacoby 2NT), and only the Generalist stops in 3NT. Where the rules decide, the profiles leave little trace; where they do not, the profile decides. That is the intended behavior of an agent built as rules first, style second.

Table 6: Same hands, five agents (final prompts). Where the rules leave room (left), the call tracks the profile; where they do not (right), the agents converge.

Agent	12 HCP, after RHO's 1♡	22 HCP, after partner's 1♠
Slam Hunter	1♠	2NT (Jacoby)
Fighter	1♠	2NT (Jacoby)
NT Specialist	1♠	2NT (Jacoby)
Generalist	Pass	3NT
Insurance Player	Pass	2NT (Jacoby)

An expert pass over the *first* outputs on these hands had exposed bridge-mechanics errors: on the slam hand the Slam Hunter announced a "splinter" (a bid that promises a singleton) while holding a doubleton, and on the competitive hand an agent overcalled 1NT on 12 HCP. We hardened the prompts with three rule blocks (convention correctness, seat awareness, self-count of HCP) and re-ran. The convention-level errors disappeared; the price is the convergence

visible on the slam hand, where the hardened agents all find the same correct call. Both prompt versions and all outputs are in the repository.

A note on the *reproducibility of LLM agents* is due here. Even at temperature 0 the API is not bit-reproducible (parallel floating-point reduction flips borderline decisions). We therefore (a) persist every raw model output, so all analyses are exactly reproducible from the archived JSON even if regeneration differs; (b) report distributions over many decisions, never single calls; and (c) pin the model version (`gemini-2.5-flash`, still offered by the provider at the time of writing). Providers retire model versions, so the archived outputs, not the model, are the reproducibility anchor: every downstream number in this paper reproduces from the archive without any model call, and re-running the LLM stages with a successor model tests distributional, not exact, reproducibility. We flag this as a general methodological requirement for LLM-agent experiments: *sample-level, not call-level, reproducibility.*

7 Stage 4: Dual-Domain Evaluation

Both domains use *fixed, deterministic* counterparts, so that all between-agent variation is attributable to the profiles: a counterpart that is the same for every agent can scale the agents' behavior but cannot create differences between them.

7.1 Bridge: Metric Falsification and a Real-Data Skill Measure

Our first bridge score rewarded bidding the level implied by combined high-card points. We falsified it with a **ZI-C–style baseline** in the spirit of Gode and Sunder's zero-intelligence constrained traders [13]: a bidder drawing uniformly from the legal calls *beat the elite-derived agents* on this metric: the score rewarded stating the formula, not making the contract. A metric that a constrained random agent can top deserves scrutiny; here the scrutiny showed that the score measured formula-following rather than contract-making, and we discarded it.

The corrected measure uses the duplicate structure of the data itself (no simulation). Every board is played at two tables by different pairs holding the same cards. A pair's result on a board is the conventional team-of-four comparison: its score minus the score of the pair that held the same cards at the other table, converted to IMPs by the official scale, which compresses large swings. Scores are taken from the pair's own perspective, so a positive value means the pair did better than its counterpart on identical cards and zero means the two tables tied. A player's skill is the mean over the boards they played, a profile's skill the mean over its players. Defensive excellence, including successful penalty doubles, scores automatically. (A raw-points variant replaces the IMP conversion by the plain point difference from the two-table mean, that is, half the cross-table difference; Table 7 reports both.) Two reference points are computed on the same 2,000 boards with published cards and scored in exactly the same way, against each real table and with the board's actual vulnerability: a ZI-C bidder that declares a uniformly random contract (any level from 1 to 7, any strain) and is scored at double-dummy, and a double-dummy oracle [1] that declares the best contract available to the pair with full knowledge of all four hands, or passes the board out when no contract makes. On this scale the ZI-C bidder sits at -8.0 IMP per board, elite players near 0, and the oracle at $+1.8$ (Fig. 2); the metric orders random $<$ human $<$ omniscient, as a skill measure must. This ordering is a necessary check on the metric, not a proof of its validity.

7.2 Negotiation: Protocol, Calibration, and a Trivial Example First

Each agent, acting as *buyer*, negotiates four business scenarios (acquiring a SaaS startup; buying out a co-founder's equity; acquiring a competitor's customer book; and a multi-year enterprise licence), three runs each, against one fixed rule-based seller. The seller opens at

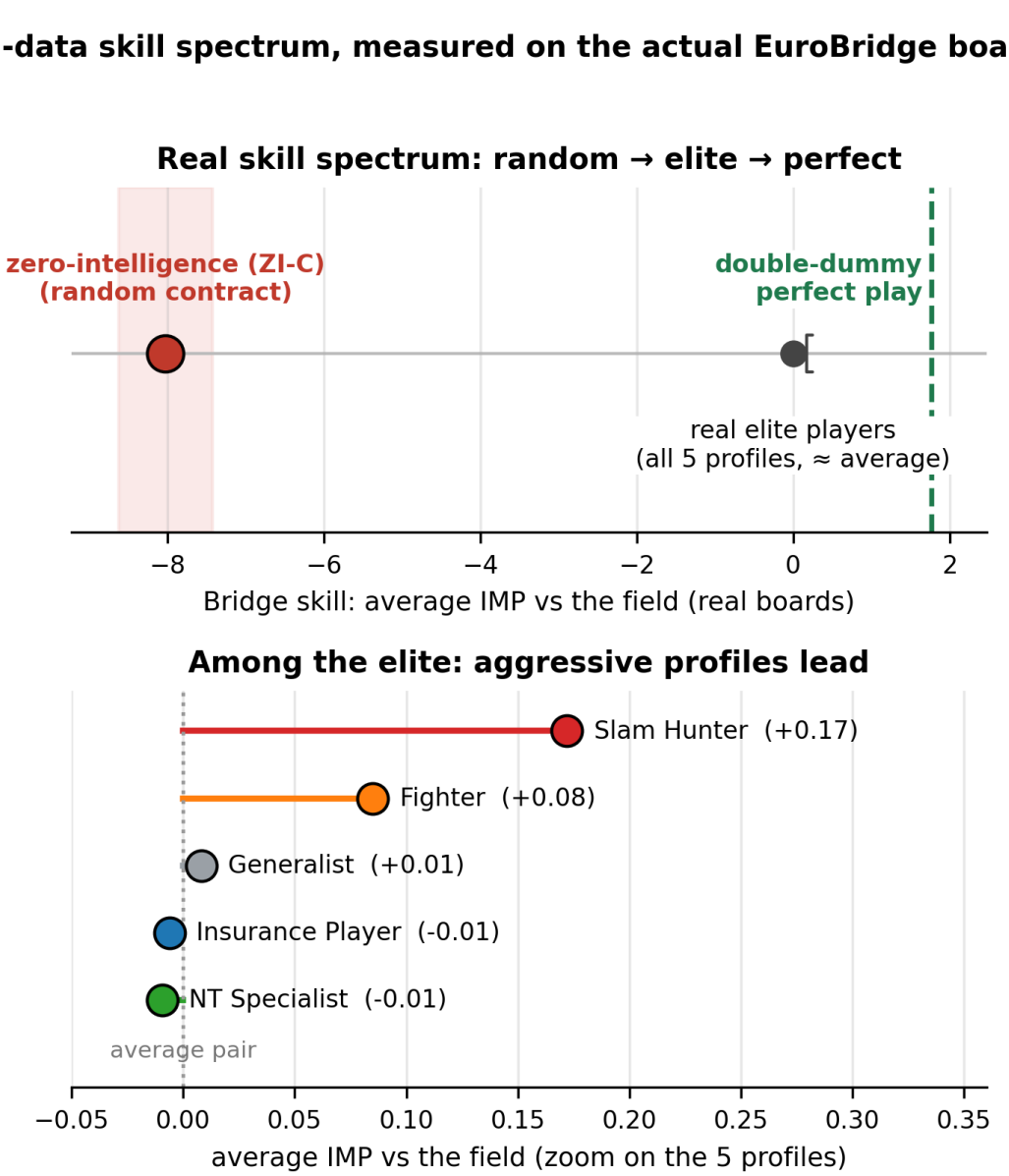


Figure 2: The skill spectrum used to check the metric: the ZI-C random baseline loses about 8 IMP per board to the real tables; elite profiles cluster near zero; the double-dummy oracle gains +1.8. A metric that cannot reproduce this ordering is measuring the wrong thing.

its ask P_{ask}, accepts any offer $\geq 0.95\,P_{\text{ask}}$, otherwise concedes 15% of the current gap per round, never below its floor P_{floor}, for at most 4 rounds; it walks away on the second "absurd" offer (below $0.70\,P_{\text{floor}}$). The *shape* of this rule, tolerance of hard bargaining with walk-away reserved for absurd offers, is grounded in **5,247 human–human bargaining dialogues** of the CraigslistBargain corpus [7]: crowdsourced negotiations between people over real Craigslist listings (the corpus' training split; 6,682 dialogues in total, of which 4,334 carry usable price fields). In that corpus aggressive lowballing captures more surplus while the deal rate barely drops ($\approx$ 0.83 vs 0.86): human sellers tolerate hard bargaining and abandon only absurd offers. The numerical constants themselves (acceptance at 95% of the ask, a 15% concession of the remaining gap, the 0.70 floor multiple, two absurd offers, four rounds) are design choices held fixed for every agent; they are not estimated from the corpus, which concerns second-hand consumer goods rather than corporate acquisitions. We rely on the corpus only for the qualitative shape of seller tolerance, and all scenario quantities below are ratios, so the absolute

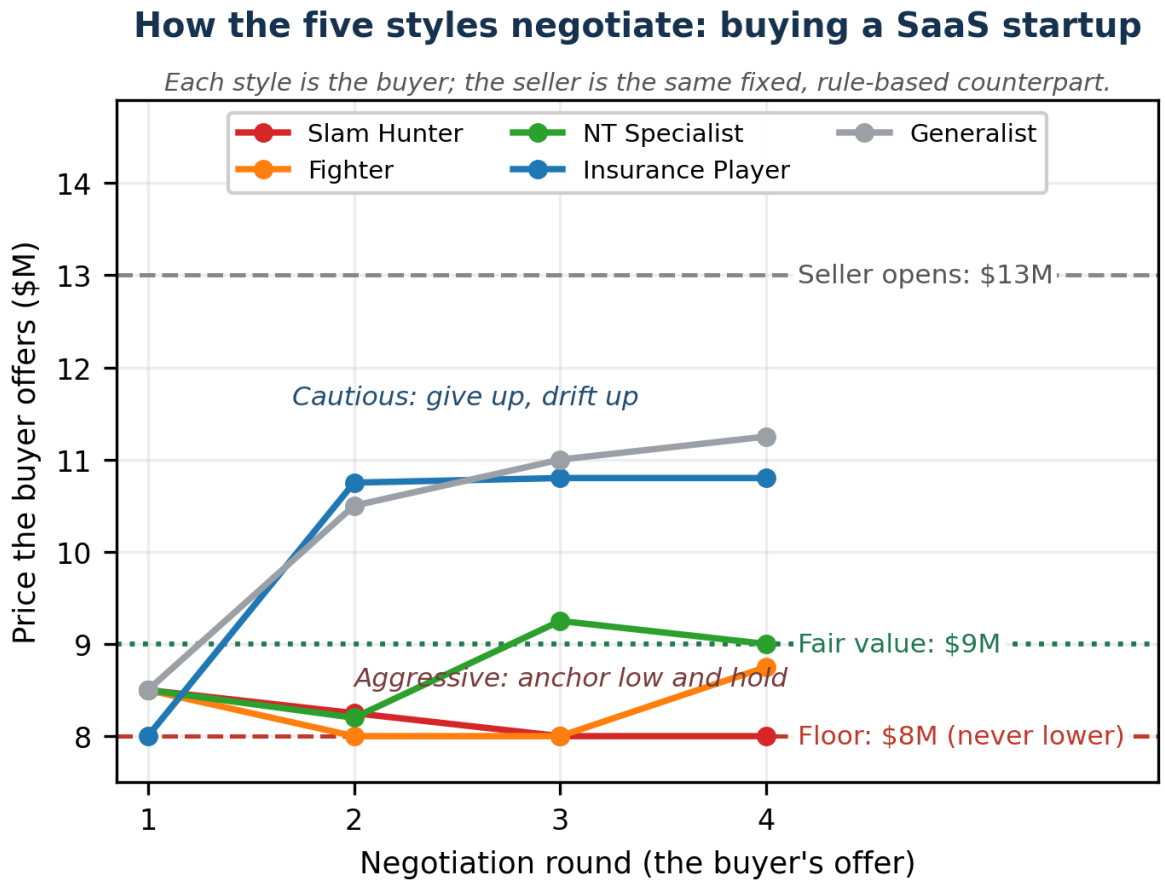


Figure 3: One negotiation, five buyers (SaaS scenario: ask \$13M, fair value \$9M, floor \$8M; real run-0 offer trajectories). Aggressive bridge profiles anchor low and hold; cautious profiles concede upward.

price scale of a scenario plays no role. An earlier, harsher walk-away rule reversed the results by punishing all aggression; we document that sensitivity in Sec. 9.1. The negotiation outcome is the *surplus captured*, $\sigma = (P_{\text{ask}} - P_{\text{deal}})/(P_{\text{ask}} - P_{\text{floor}}) \in [0, 1]$, zero if no deal. All four scenarios are single-issue price negotiations, *distributive* bargaining in the classical sense of Walton and McKersie [34]; integrative, multi-issue scenarios are deferred to future work (Sec. 11).

Following the principle of trivial-first evaluation, note the reader can predict this instrument's outcome: against a seller who concedes only under pressure, aggressive anchor-low profiles should capture more surplus, and cautious profiles should close early at worse prices. Fig. 3 shows one real scenario: the aggressive profiles anchor near the floor and hold; the cautious profiles drift up toward the seller. The instrument behaves as expected, which is what licenses its use on the non-trivial cross-domain question.

8 Implementation and Software Engineering

The pipeline is engineered as a single research software product rather than a collection of scripts, a choice worth stating explicitly given this volume's software-tools theme. All public operations flow through one SDK entry point (`src/sdk.py`); no function lives only in a notebook. LLM access is routed through a provider-agnostic client (`src/shared/llm_client.py`) that logs model, token counts, latency, and cost for every call and enforces a budget guard (alert at \$20, hard cap at \$50); swapping the model provider is a one-argument change, which is how the cross-provider replication planned as future work is already wired. The code is typed,

linted, and documented, with data hygiene enforced by convention: raw inputs are read-only, and all derived artifacts live under separate processed and results trees. A pytest suite of 178 tests covers the feature engineering, the profiling gates, the statistical validators, and the agents; LLM-dependent components are tested by injecting a mocked client, so the full suite runs without network access or cost. These practices are what make the reproduction protocol of Appendix A executable by a stranger rather than aspirational.

9 Results

9.1 Winning Shows No Stable Cross-Domain Relationship

We first correlated, per profile, the *winning* record of the profile's human players in bridge (the real-data measure of Sec. 7.1) with the winning of the profile's agent in negotiation (Spearman rank correlation [23] over the $K = 5$ profiles). This asks whether the human success of a profile's players predicts its agent's success; it does not test whether one and the same agent wins in both domains, since our agents do not play complete deals against the field. The answer depends on the bridge metric (Table 7): the discarded par-proxy gives $\rho_S = +0.30$; a fight-aware blend rises monotonically with the weight on doubling; the real-data IMP metric that passed the baseline check gives $\rho_S = +0.20$, with $+0.50$ under a raw-points robustness variant. With $K = 5$, none is significant.

Table 7: Winning↔winning is metric-sensitive. The final row is the metric of Sec. 7.1 that passed the baseline check.

Bridge-side metric	ρ_S (win vs. win, $K = 5$)
Par-level proxy (falsified by ZI-C)	$+0.30$
Fight-aware blend, weight $w = 0.2/0.3/0.4$	$+0.60$ / $+0.70$ / $+0.90$
Real-data table-vs-table IMP (passed the baseline check)	$+\mathbf{0.20}$

We read this as a substantive lesson, not a failure: *whether a style wins depends on the opponent and the payoff rules*: an over-harsh seller inverted the sign entirely in one ablation, so winning↔winning is the wrong invariant to chase across domains.

9.2 Behavioral Consistency of the Agents: $\rho_S = +0.80$

The research question was always *behavioral* alignment. Three terms are used with fixed meanings from here on: *aggression* is the bridge-side composite defined next, *demand depth* is the negotiation-side measure, and *style* is the pattern the two jointly describe; risk-taking is used only informally. Define the bridge-side aggression of a profile as the mean composite z-score of its members on the three action axes {`slam_rate`, `preempt_rate`, `penalty_double_rate`} (each z-scored across the five profile means), and the demand depth of an offer as how far below the ask it reaches,

$$\theta \;=\; \frac{P_{\text{ask}} - P_{\text{offer}}}{P_{\text{ask}} - P_{\text{floor}}}\,, \tag{2}$$

so that $\theta = 0$ is an offer at the ask and $\theta = 1$ an offer at the seller's floor. A profile's θ is the mean over every counter-offer its agent made in its 12 negotiations (four scenarios, three runs, up to four offers each), computed from the archived logs with no new simulation. First-offer anchoring is well documented [33]; here, however, the opening offers of all five agents are uniformly aggressive (θ of the first offer between 0.85 and 0.95), and using the opening offer alone gives only $\rho_S = +0.40$. The profiles separate in how the agents hold or concede over the subsequent rounds, which is what the all-offer mean captures; we report both so that the reader can see the dependence on this choice.

Table 8: Behavioral consistency across domains. θ is the mean over all offers, with the standard deviation across the profile's 12 negotiations.

Profile	Bridge aggression (z)	Negotiation demand depth θ (mean ± SD)
Fighter	+1.30	0.808 ± 0.114
Slam Hunter	+0.51	0.760 ± 0.145
NT Specialist	−0.36	0.771 ± 0.085
Generalist	−0.69	0.666 ± 0.074
Insurance Player	−0.77	0.666 ± 0.078
Spearman $\rho_S = +\mathbf{0.80}$ (exact two-sided permutation $p = 0.133$, $K = 5$)		

Agents of aggressive bridge profiles bargain more aggressively; agents of cautious ones bargain more softly (Fig. 4). The one rank inversion (NT Specialist above Slam Hunter on θ) is what caps ρ_S at 0.80 rather than 1.0; this is unsurprising, since the NT axis is stylistic rather than directly risk-graded. The between-profile differences in θ are of the same order as the within-profile standard deviations across negotiations, which is another reason to read the ranking as an indication rather than a precise estimate.

A note on inference is due here rather than in the limitations. With $K = 5$, the permutation null admits only $5! = 120$ orderings of one variable against the other. Enumerating all of them (ties handled by average ranks; the script is in the repository) gives the exact two-sided p-values: 0.017 for a perfect ranking ($|\rho_S| = 1$), 0.083 for $|\rho_S| = 0.9$, and 0.133 for the observed $|\rho_S| = 0.8$. The asymptotic approximation that statistical software reports by default ($p = 0.10$ and 0.037 for the two correlations of this paper) is not valid at this sample size, and an earlier draft of this paper used it. No correlation in this paper is therefore statistically significant on its own, including the inverse-prompt control of Sec. 9.3. We read +0.80 and −0.90 as effect-size indications whose weight comes from the pattern rather than from a p-value: a near-perfect ranking under the matched skills that reverses under swapped skills while every label stays fixed. Raising K (finer tail slices, per-player agents) is the direct fix.

9.3 The Inverse-Prompt Control: $\rho_S = -0.90$

Could the agents simply be acting out their names? Control: keep every identity line intact but inject the *opposite* profile's Stage-2 skills (Fighter gets the Insurance skill list and vice versa), then re-measure θ. If behavior follows the label, nothing should change; if it follows the skills, the correlation should invert. It inverts: the Fighter with Insurance skills softens (θ : 0.808 → 0.725), the Insurance Player with Fighter skills hardens (0.666 → 0.805), and the cross-domain correlation flips to $\rho_S = \mathbf{-0.90}$ (exact two-sided $p = 0.083$; Fig. 5). One caveat belongs here: the matched runs used `gemini-2.5-flash` and the control runs `gemini-2.5-flash-lite`, a lighter variant of the same model (Appendix A.1), so the comparison spans two model variants as well as the skill swap; a re-run of both conditions on one model is the first check listed in Sec. 10. The alignment is therefore mediated by the skill content: swapping it reverses the ranking while every label stays fixed. The control shows that the skill descriptions drive the measured behavior; it does not show that descriptions extracted from real hands are *necessary* for it. The conditions that would test necessity, labels alone, generic instructions to be aggressive or cautious, and randomized skill lists, were not run; they are the first item of future work (Sec. 10).

10 Discussion

The two findings are two sides of one coin. In our agents, decision *style*, how much is demanded and how firmly a position is held, is consistent across the two domains (+0.80, with the skill-

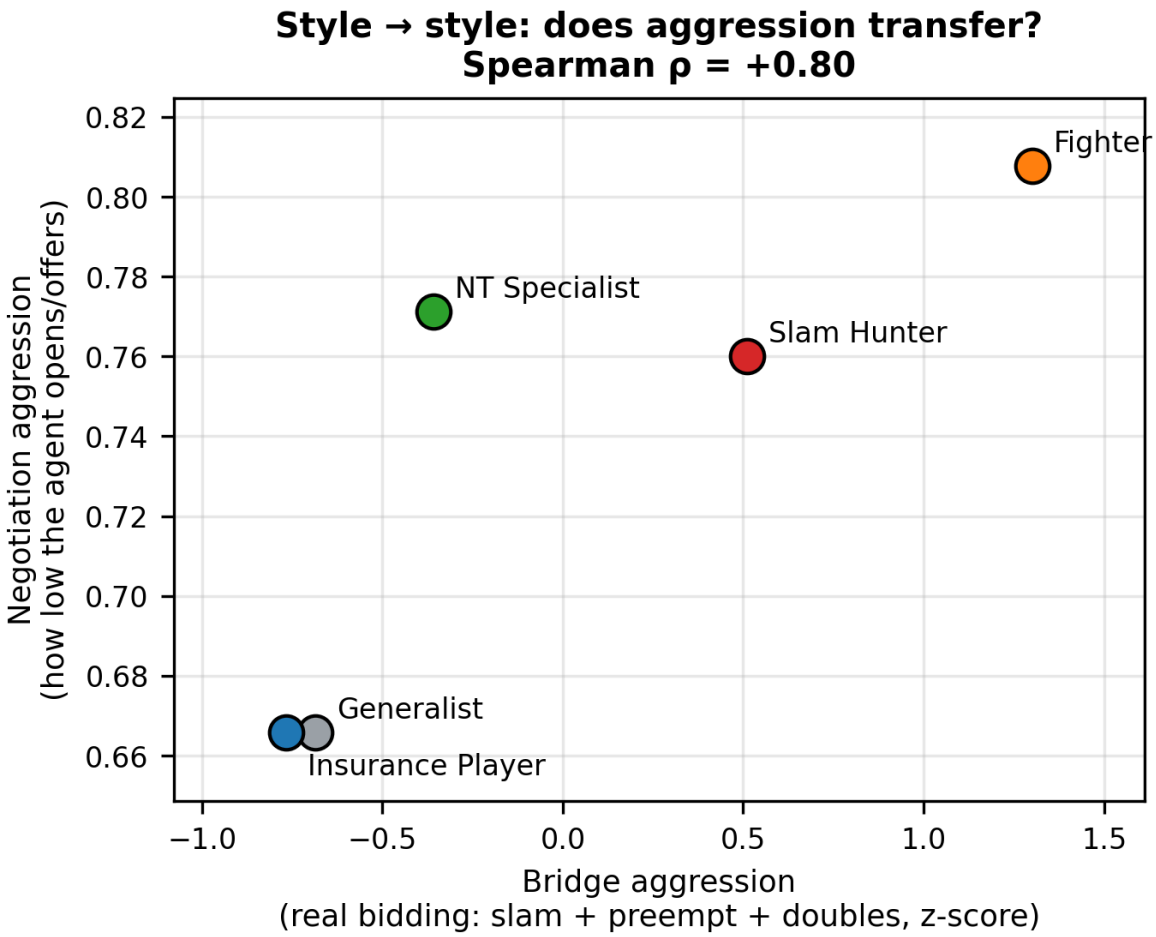


Figure 4: Behavioral alignment: bridge-derived aggression (x) vs. negotiation demand depth (y), $\rho_S = +0.80$.

swap control). *Winning* is a property of the decision-maker *jointly with* the environment's payoff structure and counterpart, and shows no stable relationship across the domains (+0.2 to +0.5, sign-reversible under an unrealistic counterpart). In optimization terms: the agents carry a stable preference structure over the risk–reward trade-off into a new decision problem, while the *value* of that preference structure depends on the objective landscape it meets. If the pattern holds for humans, which this study does not test, profiling a counterpart's style from observable past behavior would be informative about *how* they negotiate rather than about who wins.

What do these findings mean beyond the laboratory? First, the clustering result is compatible with a *dimensional* rather than typological picture of elite decision-making style: expert players appear to differ by degree along a few interpretable axes, not by membership in discrete types. This parallels the move in personality psychology from types to traits. The out-of-sample test adds the necessary caution: an individual's position on those axes is measured with low reliability when the behavior is rare and the record per person is a few hundred decisions, so behavioral profiling in other expert domains should be built on graded axes with explicit significance gates rather than on clustering, and then tested out of sample, since in-sample separation is guaranteed by the selection itself (Sec. 4.4). Second, the consistency result says that what persists across domains, in the agents, is the *shape* of the risk preference, how much is demanded and how firmly a position is held, rather than its payoff. If the same holds for people, which remains to be tested, the useful direction for decision support is clear: a counterpart's

Style transfers, and the skills carry it (anti-tautology control)

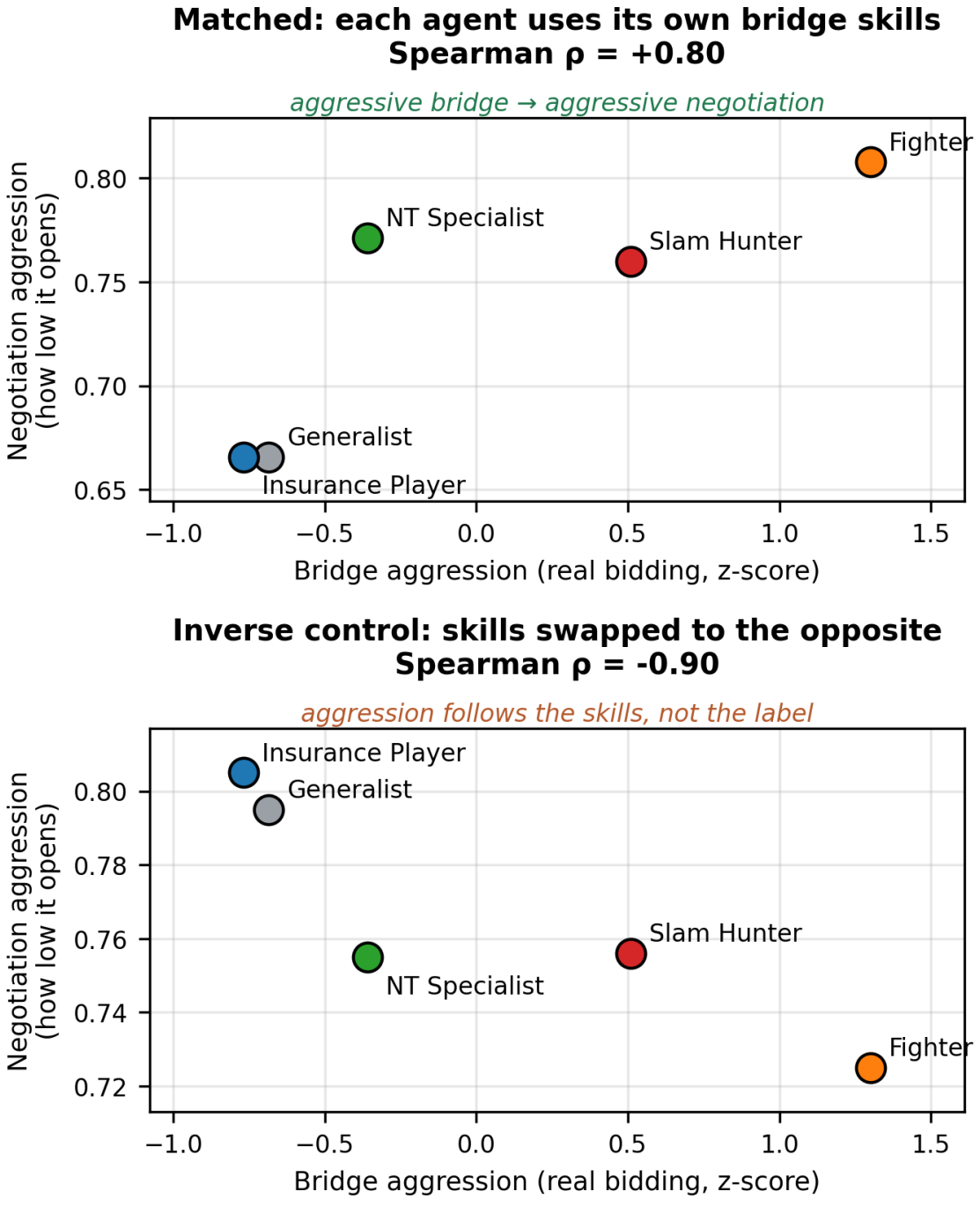


Figure 5: Inverse-prompt control. Swapping each profile's bridge-derived skills (labels unchanged) flips the alignment from +0.80 to −0.90: the behavior follows the skills, not the label.

past behavior in a structured, recorded domain would be informative about how they open and concede, even when it says little about who will win. Third, the control result shows that the behavior of a language-model agent followed the behavioral content it was given rather than the profile label; whether content extracted from real records is necessary, as opposed to any sufficiently specific behavioral instruction, remains to be tested. That is both a methodological warning and an opportunity: persona-based simulations are only as good as the evidence behind the persona.

Three components are *methodological exports*, reusable beyond bridge: (i) the significance-gated tail-profiling recipe for low-base-rate behaviors (Eq. 1), together with the out-of-sample test that must accompany it; (ii) metric falsification by constrained-random baseline (a ZI-C agent must rank last, an oracle first) before any cross-domain claim; (iii) evidence-grounded skill extraction plus an inverse-prompt control as an anti-tautology standard for LLM-persona experiments.

Several *limitations* bound these conclusions. With $K = 5$ profiles no rank correlation short of a perfect one can clear $p < 0.05$ (Sec. 9.2); the observed alignment ($p = 0.133$) and its reversal under the control ($p = 0.083$) are therefore indications, and scaling the number of profiles (finer tail slices, per-player agents) is the direct fix. The profiles themselves are extreme records rather than demonstrated stable traits (Sec. 4.4), and the Insurance Player profile in particular has no out-of-sample support and depends on a proxy definition. The control experiment shows that the skill content drives the agents' behavior but not that content extracted from real hands is necessary; the conditions that would test this (labels alone, generic aggression or caution instructions, randomized skill lists) were not run. The negotiation domain is simulated and small: a rule-based counterpart with hand-set constants whose shape, not whose values, is grounded in human data, four single-issue price scenarios with three runs each, and surplus as the sole outcome; the between-profile differences in demand depth are of the same order as the within-profile variation across negotiations. Bridge-side confounds remain (partnership effects on `slam_rate`; convention systems on `nt_rate`). The skill extractor and the agents share one model family (Gemini), and the matched and control negotiation runs used two variants of it (Sec. 9.3), so provider- and version-specific behavior cannot yet be ruled out; the provider-agnostic client of Sec. 8 makes a same-model re-run of both conditions, and then cross-provider replication (Claude, GPT), a one-argument change, and these are the first robustness items of future work. Finally, elite players are a selected population; transfer from amateur play is untested. Nothing in this paper supports using such profiles to judge an individual's competence or suitability for a role, and we caution against that use: the labels describe records in one corpus, not persons.

11 Conclusions and Future Work

This paper asked whether a decision-making profile derived from a person's real bridge record persists when that profile, instantiated as an autonomous agent, negotiates in a business setting. Within the limits of a five-profile study of simulated agents, the answer is that the *style* is consistent across the two domains and the *success* is not. Along the way the bridge record itself yielded two cautionary findings: the behavioral features of elite players show no clearly separated clusters, and the extreme profiles, although each member's record is individually distinguishable from the baseline, are only weakly stable out of sample (held-out ratios between 1.00 and 1.13; split-half reliabilities between 0.07 and 0.23). The profiles are therefore extreme records that seed the agents, not demonstrated traits of the players. The agents built from those records show a consistent risk signature across domains ($\rho_S = +0.80$; inverse control -0.90; neither significant with five profiles), while the human winning record of the profiles does not predict the agents' winning in negotiation ($\rho_S \approx +0.2$ to $+0.5$). The pipeline that produced these results, from 149,208 tournament records to agents evaluated in two domains,

is released in full so that the findings can be checked and extended.

Three extensions are planned. *Team negotiations:* multi-agent teams mixing profiles (e.g. a Fighter–Insurance buying team) to study intra-team dynamics, bridging to coalition and alliance formation. *Multi-stage deals:* a memorandum-of-understanding phase followed by substantive bargaining, testing whether style expression differs by phase. *Multi-criteria negotiation:* price plus quality plus optional features, which recasts the interaction as a multiobjective optimization problem with heterogeneous preference structures, and connects directly to interactive multiobjective decision support. On the bridge side, the natural deepening is a learned estimation–policy architecture (ENN/PNN [2]) over the full corpus, and per-player (rather than per-profile) agents to raise K from 5 to hundreds. Per-player agents are not a free lunch: the reliabilities of Sec. 4.4 show that a few hundred boards per player do not pin down an individual's rates, so such agents need records of thousands of boards per player, which online archives such as BBO Vugraph provide and championship archives do not. On the negotiation side, future work targets real human negotiation data. A concrete first step is the CaSiNo corpus [11]: 1,030 multi-issue human negotiations in which each negotiator carries measured Big-Five and social-value-orientation scores: exactly the trait→behavior linkage this paper constructs synthetically, now testable on humans (does measured trait-level aggression predict human demand depth the way bridge-derived profiles predict it in agents?). Its integrative, three-issue structure also supplies the multi-criteria scenarios for the extension above, closing the loop the laboratory opened.

Acknowledgments

The authors gratefully acknowledge the support of the Research Council of Finland through the Flagship of Advanced Mathematics for Sensing, Imaging, and Modelling (FAME), 2024–2031 (Decision No. 359183). The first author thanks Dr. Yoram Segal for the engineering-first project framework provided during the foundational AI coursework in which this system was first built. Generative AI tools (Google Gemini, Anthropic Claude) are the object of study in Stages 2–4 and were additionally used for language polishing and as coding assistants; all scientific content, design choices, analyses, and conclusions are the authors' own.

Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

References

[1] Ginsberg, M. L.: GIB: Imperfect information in a computationally challenging game. Journal of Artificial Intelligence Research **14**, 303–358 (2001)

[2] Rong, J., Qin, T., An, B.: Competitive bridge bidding with deep neural networks. arXiv preprint arXiv:1903.00900 (2019)

[3] Silver, D., et al.: A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. Science **362**(6419), 1140–1144 (2018)

[4] Kramár, J., et al.: Negotiation and honesty in artificial intelligence methods for the board game of Diplomacy. Nature Communications **13**, 7214 (2022)

[5] Meta Fundamental AI Research Diplomacy Team (FAIR), Bakhtin, A., et al.: Human-level play in the game of Diplomacy by combining language models with strategic reasoning. Science **378**(6624), 1067–1074 (2022)

[6] Lewis, M., Yarats, D., Dauphin, Y., Parikh, D., Batra, D.: Deal or no deal? End-to-end learning of negotiation dialogues. In: Proc. EMNLP 2017, pp. 2443–2453 (2017)

[7] He, H., Chen, D., Balakrishnan, A., Liang, P.: Decoupling strategy and generation in negotiation dialogues. In: Proc. EMNLP 2018, pp. 2333–2343 (2018)

[8] Vaccaro, M., Caosun, M., Ju, H., Aral, S., Curhan, J. R.: Advancing AI negotiations: A large-scale autonomous negotiation competition. Proceedings of the National Academy of Sciences **123**(23), e2521774123 (2026)

[9] Bianchi, F., Chia, P. J., Yuksekgonul, M., Tagliabue, J., Jurafsky, D., Zou, J.: How well can LLMs negotiate? NegotiationArena platform and analysis. In: Proc. ICML 2024 (2024)

[10] Abdelnabi, S., Gomaa, A., Sivaprasad, S., Schönherr, L., Fritz, M.: Cooperation, competition, and maliciousness: LLM-stakeholders interactive negotiation. In: Proc. NeurIPS 2024, Datasets and Benchmarks Track (2024)

[11] Chawla, K., Ramirez, J., Clever, R., Lucas, G., May, J., Gratch, J.: CaSiNo: A corpus of campsite negotiation dialogues for automatic negotiation systems. In: Proc. NAACL 2021, pp. 3167–3185 (2021)

[12] Mannekote, A., Davies, A., Li, G., Boyer, K. E., Zhai, C., Dorr, B. J., Pinto, F.: Do role-playing agents practice what they preach? Belief-behavior consistency in LLM-based simulations of human trust. arXiv preprint arXiv:2507.02197 (2025)

[13] Gode, D. K., Sunder, S.: Allocative efficiency of markets with zero-intelligence traders: Market as a partial substitute for individual rationality. Journal of Political Economy **101**(1), 119–137 (1993)

[14] Kahneman, D., Tversky, A.: Prospect theory: An analysis of decision under risk. Econometrica **47**(2), 263–291 (1979)

[15] Cutler, A., Breiman, L.: Archetypal analysis. Technometrics **36**(4), 338–347 (1994)

[16] Brandenburger, A. M., Nalebuff, B. J.: Co-opetition. Currency Doubleday, New York (1996)

[17] Raiffa, H.: The Art and Science of Negotiation. Harvard University Press, Cambridge, MA (1982)

[18] Nash, J. F.: The bargaining problem. Econometrica **18**(2), 155–162 (1950)

[19] Rousseeuw, P. J.: Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics **20**, 53–65 (1987)

[20] Campello, R. J. G. B., Moulavi, D., Sander, J.: Density-based clustering based on hierarchical density estimates. In: Proc. PAKDD 2013, LNCS 7819, pp. 160–172. Springer (2013)

[21] Cohen, J.: Statistical Power Analysis for the Behavioral Sciences, 2nd edn. Lawrence Erlbaum Associates, Hillsdale, NJ (1988)

[22] Mann, H. B., Whitney, D. R.: On a test of whether one of two random variables is stochastically larger than the other. Annals of Mathematical Statistics **18**(1), 50–60 (1947)

[23] Spearman, C.: The proof and measurement of association between two things. American Journal of Psychology **15**(1), 72–101 (1904)

[24] Ericsson, K. A., Lehmann, A. C.: Expert and exceptional performance: Evidence of maximal adaptation to task constraints. Annual Review of Psychology **47**, 273–305 (1996)

[25] Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., Bernstein, M. S.: Generative agents: Interactive simulacra of human behavior. In: Proc. UIST 2023 (2023)

[26] Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., Wingate, D.: Out of one, many: Using language models to simulate human samples. Political Analysis **31**(3), 337–351 (2023)

[27] Aher, G., Arriaga, R. I., Kalai, A. T.: Using large language models to simulate multiple humans and replicate human subject studies. In: Proc. ICML 2023 (2023)

[28] Virtanen, P., et al.: SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods **17**, 261–272 (2020)

[29] Pedregosa, F., et al.: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research **12**, 2825–2830 (2011)

[30] Epstein, S.: The stability of behavior: I. On predicting most of the people much of the time. Journal of Personality and Social Psychology **37**(7), 1097–1126 (1979)

[31] Mischel, W., Shoda, Y.: A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. Psychological Review **102**(2), 246–268 (1995)

[32] Weber, E. U., Blais, A.-R., Betz, N. E.: A domain-specific risk-attitude scale: Measuring risk perceptions and risk behaviors. Journal of Behavioral Decision Making **15**(4), 263–290 (2002)

[33] Galinsky, A. D., Mussweiler, T.: First offers as anchors: The role of perspective-taking and negotiator focus. Journal of Personality and Social Psychology **81**(4), 657–669 (2001)

[34] Walton, R. E., McKersie, R. B.: A Behavioral Theory of Labor Negotiations: An Analysis of a Social Interaction System. McGraw-Hill, New York (1965)

[35] Lloyd, S. P.: Least squares quantization in PCM. IEEE Transactions on Information Theory **28**(2), 129–137 (1982)

[36] Schwarz, G.: Estimating the dimension of a model. Annals of Statistics **6**(2), 461–464 (1978)

[37] Jolliffe, I. T.: Principal Component Analysis, 2nd edn. Springer, New York (2002)

[38] Clopper, C. J., Pearson, E. S.: The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika **26**(4), 404–413 (1934)

A Reproducibility of Results

This appendix collects everything a reader needs to reproduce the numbers in the paper: the fixed constants and the software environment (Sec. A.1), and a numbered, end-to-end protocol that maps each table and figure to the script that produces it and states the expected output of every step (Sec. A.2).

A.1 Fixed Constants and Environment

All code, prompts, thresholds, seeds, the processed corpus, the per-player profile table, and the archived model outputs are in the public repository:

`https://github.com/annafr2/From-Bridge-Tables-to-Global-Markets`

The fixed constants are: qualification $n \geq 50$ declared and ≥ 50 bidding boards; percentile gate top 10%; binomial gate $\alpha = 0.05$ (one-sided, SciPy `binomtest` [28]); cv-filter 0.10; K-Means/GMM/HDBSCAN via scikit-learn [29] with `random_state`=42; LLM chunking 25 boards/call, ≤ 50 boards/player, sampling seed 42; skill-merge TF–IDF threshold 0.40; temperatures 0.3 (extraction), 0.3/0.0 (bridge), 0.7 (negotiation); seller rule: accept $\geq 0.95\,P_{\text{ask}}$, concede 15% of gap/round, floor-bounded, ≤ 4 rounds, walk-away after 2 offers below $0.70\,P_{\text{floor}}$.

The environment is Python 3.11 with scikit-learn, SciPy 1.11, and pandas (versions pinned in `requirements.txt`). The LLM stages use two variants of one model family through a single provider-agnostic client: `gemini-2.5-flash` for skill extraction, the bridge agents, and the matched negotiation runs, and `gemini-2.5-flash-lite` for the inverse-prompt control runs. Every call is logged with model, token counts, latency, and cost in the released per-call log (folder `results/llm_logs/`); the total compute cost of all experiments was under $10.

A.2 Step-by-Step Protocol

The protocol below reproduces every table and figure in this paper. Steps 1–6 are fully deterministic (no API key, no cost). Steps 7–8 and 11 call the LLM and are therefore *distributionally* reproducible (Sec. 6); the raw model outputs of our runs ship in the repository, so steps 9–10 and 12 can be reproduced *exactly* from the archive without re-querying any model. Every command is run from the root of the `NegoPlay` repository; the version of the code that produced this paper is the Git tag `paper1-op80-final`.

1. **Environment.** Clone the repository (URL in Sec. A.1); create a Python 3.11 virtual environment; run `pip install -r requirements.txt`. For steps 7–8 and 11 only, set `GOOGLE_API_KEY` in `.env`.

2. **Data check.** The corpus and the per-player profile table ship with the repository, in the folder `data/processed/` (the corpus one level above the `NegoPlay` folder, 2 MB):
   `all_matches.parquet`
   `player_profiles.csv`
   Load the corpus via `src/shared/data_loader.py`.
   Expected: 149,208 board-room records; per-championship coverage as in Table 1.

3. **Stage 1: features and profiles.**
   Run the Stage-1 pipeline in `src/stage1_clustering/`: first `features.py` (with `MIN_BOARDS` = 50), then `extreme_profiles.py` (top-10% cutoff, $\alpha = 0.05$).
   Expected: `data/processed/player_profiles.csv` with 563 rows and profile counts 20/21/37/17/468 (Table 4).

4. **Cluster-structure audit.**
   `python -m notebooks.preprocessing_comparison`
   Expected: the five configurations of Table 3 (max silhouette 0.24; HDBSCAN 0 clusters) and `results/preprocessing.xlsx`.

5. **In-sample profile check.**
   `python -m notebooks.validate_base_rates`
   Expected: the in-sample separations mentioned in Sec. 4.4 (Cohen's d 2.1–4.6), which the paper does not use as evidence.

6. **Out-of-sample validation and sensitivity.**
   `python -m notebooks.split_half_validation 50 match`
   `python -m notebooks.feature_definition_sensitivity`
   Expected: Table 5, the split-half reliabilities and membership-stability figures of Sec. 4.4, and the strict-definition overlaps of Sec. 4.4 (Jaccard overlaps of 1.00, 0.14, 0.95, and 0.94, and $\rho_S = +0.80$ under the strict preempt definition). Passing `parity` instead of `match` reproduces the dealer-rotation artefact discussed in the text.

7. **Stage 2: skill extraction** (LLM; ≈$1).
   Run the extractor in `src/stage2_skills/` over the Stage-2 sample: five players drawn at random (seed 42) from each of the four extreme profiles, 20 players in all; chunks of 25 boards, at most 50 boards per player, temperature 0.3, and the model `gemini-2.5-flash`; then aggregate with the TF–IDF merge threshold 0.40.
   Output: the per-profile skill signatures JSON; the archived run used in the paper is `results/stage2_sample_v2_focused_prompt/`.

8. **Stages 3–4: dual-domain simulation** (LLM; <$5).
   `python -m src.stage4_simulate.bridge_runner`

`python -m src.stage4_simulate.negotiation`
All raw agent outputs are persisted under `results/stage4/`.

9. **Bridge skill metric and spectrum.**
`python -m notebooks.alignment_real_bridge`
`python -m notebooks.visualize_real_skill_spectrum`
Expected: Fig. 2 and the win–win row of Table 7 ($\rho_S = +0.20$ on IMPs, $+0.50$ on raw points).

10. **Style alignment.**
`python -m notebooks.style_alignment`
Recomputed from the archived Stage-4 logs with no new simulation. Expected, *exactly*: Table 8 and Fig. 4 ($\rho_S = +0.80$).

11. **Inverse-prompt control** (LLM; <$1).
`python -m notebooks.inverse_prompt_control`
Re-runs the swapped-skill negotiations with live model calls, so it reproduces distributionally; the archived outputs behind Fig. 5 ($\rho_S = -0.90$) ship in the repository.

12. **Seller calibration.**
`python -m notebooks.validate_negotiation_features`
Runs against the CraigslistBargain training split under `data/external/`. Expected: 4,334 usable dialogues; aggressive opening captures more surplus at a deal rate of ≈ 0.83 vs 0.86 (Sec. 7.2).